

RAYMOND R. WAYESU

Data Analyst / Biostatistician

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PROFESSIONAL SUMMARY

Medical Statistician and Data Analyst with over 10 years of experience in biostatistics, health data quality assurance, and monitoring and evaluation in Uganda and internationally. Currently serving as Medical Statistician at the Uganda Virus Research Institute (UVRI) / London School of Hygiene & Tropical Medicine (LSHTM). Holds an MSc in Mathematical Statistics from Linköping University, Sweden, and a BSc in Statistics from Makerere University. Demonstrated expertise in routine data quality assessment, HMIS data verification, digital data collection instrument development, statistical modelling, and dashboard development. Proficient in R, Python, SAS, STATA, and SQL. GCP certified. Experienced in working with health facility data, clinical trial data quality control, and national survey data management.

KEY COMPETENCIES FOR RDQA CONSULTANCY

- Data Quality Assessment: Routine data quality audits, data verification and validation, Verification Factor computation, three-way data triangulation (facility registers, digital platforms, DHIS2)
- Health Data Systems: HMIS register review (031, 055B, 071, 074, 080/081), DHIS2 data extraction and analysis, WHO Data Quality Review Toolkit, MEASURE Evaluation RDQA Tool
- Digital Data Collection: KoboToolbox/KoboCollect instrument design and deployment, REDCap, ODK, electronic data capture system management with built-in validation rules
- Statistical Analysis: Biostatistical modelling, sample size estimation, trend analysis, facility performance scorecards, spider chart generation, root cause analysis
- Programming & Tools: R (tidyverse, Shiny, ggplot2, plotly), Python (pandas, NumPy, scikit-learn), SAS, STATA, SQL, SPSS, LaTeX, Linux/Unix, Git/GitHub
- Dashboard & Reporting: R/Shiny interactive dashboards, data visualisation, automated report generation, Power BI, facility-level scorecards
- Research & Quality: GCP certified (2023), clinical trial data management, manuscript preparation, peer-reviewed publications, ethical research practices

PROFESSIONAL EXPERIENCE

Medical Statistician

Uganda Virus Research Institute (UVRI) / London School of Hygiene & Tropical Medicine (LSHTM) | 2024 – Present

Coalition for Epidemic Preparedness Innovations (CEPI) – Immunology & Vaccine Studies

- Lead statistical analysis for immunology studies, including research design, sample size estimation, and power calculations for vaccine efficacy trials.
- Ensure data quality and integrity of laboratory datasets through systematic quality assurance protocols, identifying and resolving data discrepancies.
- Develop and maintain statistical analysis plans (SAPs) using R and SAS, ensuring reproducibility and regulatory compliance.
- Manage and clean large immunological datasets, performing validation checks and producing analysis-ready datasets.

- Prepare statistical reports, tables, and figures for manuscripts submitted to peer-reviewed journals.
- Collaborate with multidisciplinary teams including immunologists, clinicians, and epidemiologists across international research networks.

Hub Data Supervisor

Sanyu Africa Research Institute – BabyGel Cluster-Controlled Trial | 2023 – 2024

72 villages across Eastern Uganda (Jinja, Iganga, Kamuli)

- Supervised data quality across 72 village clusters, ensuring accuracy and completeness of health facility and community-level data collection.
- Conducted routine data quality checks on electronically captured data, identifying discrepancies between field reports and database entries.
- Trained and mentored hub data collectors on data quality protocols, digital data entry, and standard operating procedures.
- Managed data flow from field collection to central database, implementing validation rules and data cleaning procedures.
- Produced weekly data quality reports for the principal investigator, tracking enrolment, completeness, and error rates.
- Coordinated with facility-level health workers and community health teams for participant follow-up and data verification.

Statistician

National Social Security Fund (NSSF) Uganda | 2022 – 2023

- Performed data quality assurance and validation on member contribution records, identifying inconsistencies and ensuring data integrity.
- Developed statistical models for actuarial analysis and member benefit projections using R and Python.
- Built automated data quality dashboards and reporting tools, reducing manual data review time.
- Conducted trend analysis on contribution patterns and compliance metrics across employer categories.

Research Assistant / Data Analyst

Makerere University – iTECH Maternal Health Project | 2021 – 2024

- Contributed to data quality assessment and statistical analysis for maternal health outcomes research.
- Developed prediction models for preterm birth and maternal complications using machine learning techniques in R and Python.
- Performed data cleaning, validation, and transformation of facility-level health data for publication-ready analysis.
- Co-authored manuscripts for peer-reviewed journals based on statistical findings.

Research Assistant

Iganga-Mayuge Health and Demographic Surveillance Site (HDSS), MakSPH | 2017 – 2019

- Conducted household health surveys and routine data quality verification for the demographic surveillance system.
- Managed field data collection teams, ensuring adherence to data quality protocols and standard operating procedures.
- Performed data entry, cleaning, and validation using STATA and SPSS, maintaining the longitudinal HDSS database.
- Participated in community engagement activities and health facility data verification exercises.

Data Processor

Uganda Bureau of Statistics (UBOS) | 2018 – 2019

- Processed and validated national survey data, ensuring data quality and consistency across survey instruments.
- Applied data cleaning algorithms and validation checks to large-scale population datasets.
- Generated statistical summaries and cross-tabulations for national reporting.

Research Assistant

Makerere University School of Public Health | 2016 – 2017

- Supported health facility data collection and quality verification for public health research projects.
- Assisted in statistical analysis using STATA and data management for epidemiological studies.

EDUCATION

MSc Mathematical Statistics

Linköping University, Sweden | 2019 – 2021

BSc Statistics

Makerere University, Kampala, Uganda | 2012 – 2015

CERTIFICATIONS & TRAINING

- ▶ Good Clinical Practice (GCP) Certification – 2023
- ▶ Advanced Statistical Methods for Clinical Trials – Linköping University
- ▶ Machine Learning and Data Science – Self-directed (Python, scikit-learn, TensorFlow)

TECHNICAL SKILLS

- ▶ Statistical Software: R (advanced – Shiny, tidyverse, ggplot2, plotly), Python (pandas, NumPy, scikit-learn), SAS, STATA, SPSS
- ▶ Data Management: SQL, REDCap, KoboToolbox/KoboCollect, ODK, Excel (advanced)
- ▶ Programming: R, Python, SQL, LaTeX, Bash/Shell scripting, Git/GitHub
- ▶ Operating Systems: Linux/Unix (proficient), Windows, macOS
- ▶ Visualisation: R/Shiny dashboards, ggplot2, plotly, D3.js, Power BI
- ▶ Health Data Systems: DHIS2, HMIS tools, WHO Data Quality Review App
- ▶ Languages: English (fluent), Luganda (native), Swedish (basic)

PORTFOLIO: RDQA TOOLS DEVELOPED FOR THIS CONSULTANCY

In preparation for this assignment, I have developed two complementary digital tools that demonstrate our team's technical capabilities and analytical approach:

- ▶ RDQA Demo Dashboard (rdqa-demo.vercel.app): A static Next.js website showcasing Verification Factor analysis, systems assessment spider charts, root cause analysis, and three-way data triangulation with interactive D3.js visualisations.
- ▶ RDQA Web Tool (dnfxdz-raymond-wayesu.shinyapps.io/rdqa-web-tool/): A fully functional R/Shiny application providing interactive data entry, real-time Verification Factor computation, M&E systems assessment scoring, DATIM validation checks, and automated PDF report generation — with both demo mode and live DHIS2 integration capability.

SELECTED PUBLICATIONS

- ▶ Mubiri P, Wayesu R, et al. Prediction of preterm birth using machine learning techniques: a systematic review. (iTECH Project, Makerere University).
- ▶ Contributions to peer-reviewed publications in maternal health, immunology, and health systems research through UVRI/LSHTM, Sanyu Africa, and Makerere University collaborations.
- ▶ Full publication list available upon request.

REFERENCES

Available upon request.